A GUIDE TO PRODUCT QUALITY STANDARDS

MISCELLANEOUS PRODUCTS

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INDEX

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ABBREVATIONS

ACCC	Australian Competition and Consumer Commission
ANSI	American National Standards Institute
APLAC	Asia Pacific Laboratory Accreditation Co-operation
APHIS	Animal and Plant Health Inspection Service
ARTG	Australian Register of Therapeutic Goods
AS	Australian Standards
ASTM	American Society for Testing and Materials
BICON	Bio Security Import Condition System
BIS	Bureau of Indian Standards
CE	Conformité Européenne
CoC	Certificate of Conformity
CPSC	Consumer Products Safety Commission
CPSIA	Consumer Product Safety Improvement Act
CPSA	Consumer Product Safety Act
DCL	Dubai Central Laboratory Department
DOP	Di-Octyl Phthalate
EC	Council Regulation
ECHA	European Chemicals Agency
EFEO	European Federation of Essential Oils
EN	European Standards
EPA	Environment Protection Act
EU	European Union
EUTR	European Timber Regulation
FCM	Food Contact Materials
FDA	Food and Drug Administration
FHSA	Federal Hazardous Substances Act
FSSAI	Food Safety and Standards Authority of India
GCC	Gulf Co-operation Council
GI	Geographical Indication
GSO	GCC Standardization Organization
GPSD	General Product Safety Directive
IFRA	International Fragrance Association
ILAC	International Laboratory Accreditation Co-operation
IS	Indian Standard
ISI	Indian Standards Institution
ISO	International Organization for Standardization
MDF	Medium Density Fibre
MLA	Multilateral Recognition Arrangement
MRA	Mutual Recognition Arrangement
NABCB	National Accreditation Board for Certification Bodies
NABL	National Accreditation Board for Testing and Calibration Laboratories
REACH	Registration, Evaluation, Authorization and Restriction of Chemicals
RoHS	Restrictions on Hazardous Substances Directive
SASO	Saudi Standards, Metrology and Quality Organization
TGA	Therapeutic Goods Act
UAE	United Arab Emirates
QCI	Quality Council of India
QCO	Quality Control Order

BACKGROUND

Uttar Pradesh, the Indian state with a geographical expanse of 2, 40,928 sq.km and a population of 204.2 million people is home to a great diversity in all facets of life. With diverse community traditions and economic pursuits, Uttar Pradesh is also endowed with a beautiful diversity of crafts and industries spread across small towns and districts known for the uniqueness of these products.

To encourage such indigenous products and crafts, the Uttar Pradesh government has initiated the One District One Product (ODOP) Scheme with a vision to create product-specific traditional industrial hubs across 75 districts of Uttar Pradesh that will also promote traditional industries that are synonymous with the respective districts of the state. Although many of the district-specific industries are more of a commonplace but there are several products that are unique to those regions.

Many of these products are GI-tagged; a geographical indication (GI) is a sign used on products that have a specific geographical origin and possess qualities or a reputation that are due to that origin. This becomes a testimony to the original creativity and exclusiveness of some of the finest products produced in the various districts of Uttar Pradesh.

To help in achieving the objectives of the ODOP initiative, the ODOP Cell, Government of Uttar Pradesh, engaged Quality Council of India (QCI) to develop 'Compendiums of relevant Regulations and Standards' for the products under this initiative. The compendiums have been developed by the Quality Council of India with assistance of the sectoral and product experts. Field visits and virtual interactions with ODOP manufacturers were also undertaken to assess the level of their understanding on relevant standards and certifications. The ODOP Cell, Government of Uttar Pradesh and the EY team at the ODOP Cell also provided continuous support in the development of these compendiums.

The key objective of developing the compendiums is to provide a ready reference of various regulations & standards for the existing ODOP manufacturers and producers and also for the aspiring entrepreneurs.

This part of the compendium brings relevant regulations and standards pertaining to various products like agarbatti, sanitary fittings, attar, etc. manufactured in different districts that provide employment to a large number of people in the respective districts.

AGARBATTI AND DHOOPBATTI, KANNAUJ

1. INTRODUCTION

Agarbatti and dhoopbatti (incense sticks) are used for religious purposes in India and across many cultures worldwide. Kannauj district is known for the production of incense sticks. The oldest source on incense are the Vedas especially, the Atharva Veda and the Rigveda, which set out and encouraged a uniform method of making incense. Although, the Vedic texts mention the use of incense for masking odours and creating a pleasurable smell, the modern system of organized incense-making was likely created by the medicinal priests of the time. Incense sticks are aromatic biotic material that release fragrant smoke when burned. The term is used for either the material or the aroma. Incense is used for aesthetic reasons, religious worship, aromatherapy, meditation and other important ceremonies. It may also be used as a simple deodorant or insect repellent.

Incense is composed of aromatic plant materials, often combined with essential oils. The forms taken by incense differ with the underlying social culture and have changed with advances in technology and increasing number of uses. Incense can generally be separated into two main types: "indirect-burning" and "direct-burning". Indirect-burning incense (or "non-combustible incense") is not capable of burning on its own, and requires a separate heat source. Direct-burning incense (or "combustible incense") is lit directly by a flame and then fanned or blown, leaving a glowing ember that releases a smoky fragrance. Direct-burning incense is either a paste formed around a bamboo stick, or a paste that is extruded into a stick or cone shape.

Various resins, such as amber, myrrh, are used in the manufacture of traditional masala incense, usually as a fragrant binding ingredient, and these add their distinctive fragrance to the finished incense.

Dhoops are extruded incense, lacking a bamboo stick core. Many dhoops have very concentrated scents and emit a lot of smoke when burned. The most well-known dhoop is probably Sandalwood (Chandan) dhoop. It contains a high percentage of sandalwood.

2. RAW MATERIALS

The raw materials for manufacturing agarbattis and dhoops are:

- Charcoal powder
- Sawdust

- Incense powder
- Bamboo sticks
- Dhoop wood

3. MANUFACTURING PROCESS

Production may be partly or completely by hand or machine. There are both automatic and semi-automatic machines available for applying paste, perfume-dipping, packing, etc., although the bulk of production is done by hand-rolling at home.

The basic ingredients of an incense stick are bamboo sticks, paste (generally made of charcoal dust or sawdust and joss/jiggit/gum/tabu powder – an adhesive made from the bark of *litsea glutinosa* and other trees), and the perfume ingredients – which traditionally would be a masala (powder of ground ingredients), though more commonly is a solvent of perfumes and/or essential oils. After the base paste has been applied to the bamboo stick, it is either, in the traditional method, while still moist, immediately rolled into the masala, or, more commonly, left for several days to dry, and then dipped into the scented solvent. For direct burning incense, the paste is cut and dried into pellets. Incense of the floral fragrances are the most common. The incense mixture is then rolled out into a slab in desired thickness and left until the slab is firmed. It is then cut into small cubes, coated with clay powder to prevent adhesion and allowed to fully harden and dry. Certain proportions are necessary for direct-burning incense and they can be described as below:

3.1. Oil Content

An excess of oils may prevent incense from smouldering effectively and resinous materials such as myrrh are thus typically balanced with "dry" materials such as wood, bark and leaf powders.

3.2. Oxidizer Quantity

Too little oxidizer in gum-bound incense may prevent the incense from igniting, while too much will cause the incense to burn too quickly, without producing the charecteristic fragrant smoke.

3.3. Binder

Water-soluble binders are added to ensure that the incense mixture does not crumble when dry.

3.4. Mixture density

Incense mixtures made with natural binders must not be combined with too much water in mixing or over-compressed while being formed, which would result in either uneven air distribution or undesirable density in the mixture, causing the incense to burn unevenly, too slowly, or too quickly.

3.5. Particulate size

The incense mixture has to be well pulverized with similarly sized particulates. Uneven and large particulates result in uneven burning and inconsistent aroma production when burned.

To manufacture Dhoopbatti, eucalyptus leaves, tulsi leaves & other leaves as required, are cut with the help of leaf cutting machines and then dried. After drying, the leaves are pulverized and converted in powder form. Guggal, cascalia powder, jigat powder and dried compounds are mixed with diethyl pthylate (D.E.P) in water bath (mixing machine). Next, dhoop wood, charcoal powder, ground net shell powder and powder leaves are mixed uniformly in powder mixing machine. This material along with rubber solution (plasticizer) and perfumery compound is transferred in edge runner machine. The edge runner machine is a precisely engineered mixer which is used extensively in chemical, cosmetics and food industries. The material gets converted in sticky paste which is transferred in cutting machine to convert it incense sticks.

Figure 1: Incense mixer

4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- All the incense sticks should be of same size
- The sticks should not be broken or damaged
- The incense sticks should not contain moisture

5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations (domestic) and international (some key regions/countries) are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organisations (https://www.fieo.org/)
- Directorate General of Foreign Trade (https://www.dgft.gov.in/CP/)
- Export Promotion Councils (https://commerce.gov.in/useful-links/export-promotion-councils/)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (https://commerce.gov.in/international-trade/trade-agreements/).

5.1. Domestic

S. No.	Country	Regulation	Regulatory Body	Remarks
1	India	The Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006 – Minor Forest Produce	Tribal Cooperative Marketing Development Federation of India Limited	Resins used in production of agarbattis

5.2. International

S. No.	Country	Regulation	Regulatory Body	Remarks
1	European union	**a.** Registration, Evaluation, Authorization and Restriction of Chemicals (REACH) Regulations, (EC 1907/2006)	**a.** European Commission **b.** European Chemicals Agency (ECHA)	**a.** Regulations on chemical used in fragrance industry **b.** Hazardous chemicals
2	United States of America	**a.** Code of Federal Regulations, Tile 27, part 21, Subchapter D	**a.** Consumer Product Safety Commission (CPSC)	**a.** Regulations on chemicals of incense **b.** Quantity of regulated material
3	Australia	**a.** Therapeutic Goods Act, 1989	**a.** Department of Health	**a.** Regulations on hazards associated to product **b.** Regulation on chemicals used
4	Middle east	**a.** Saudi Standards, Metrology and Quality Organization (SASO) Regulations	**a.** Safety, Saudi Standards, Metrology and Quality Organization (SASO)	**a.** General product safety regulations **b.** Labelling of product

6. VOLUNTARY STANDARDS

Domestic Standards			International Standards		
Stage	**Code**	**Aspects Covered**	**Region/ Country**	**Standard**	**Aspects Covered**
General (Refer Table 1)	**IS 6597: 2001 (Reaffirmed Year: 2017)**	Glossary of terms: Fragrance and flavour	European Union	**ISO 21625: 2020**	Vocabulary of bamboo products
	-	-	Australia	**AS ISO 21625: 2020**	Vocabulary of bamboo products

Domestic Standards			International Standards		
Stage	**Code**	**Aspects Covered**	**Region/ Country**	**Standard**	**Aspects Covered**
Raw Materials (Refer Table 2)	**IS 3349: 1989 (Reaffirmed Year: 2020)**	Perfumery Materials - Resinoid benzoin, pure	-	-	-
	IS 6874: 2008 (Reaffirmed Year: 2019)	Tests for bamboo	-	-	-
	IS 326: Part 1: 2018 - IS 326: Part 2: 1918 (Reaffirmed Year: 2017) IS 326: Part 7: 2006 (Reaffirmed Year: 2017) IS 326: Part 8: 2005 (Reaffirmed Year: 2017)	Methods of sampling and test for natural and synthetic perfumery material	-	-	-
Manufacturing/ Processing (Refer Table 3a. & 3.b)	**IS 329: 2004 (Reaffirmed Year: 2020)**	Oil of sandalwood	European Union	**ISO 3518: 2022**	Oil of sandalwood
	IS 512: 1988 (Reaffirmed Year: 2019)	Oil of citronella(Java)	European Union	**ISO 3848: 2016**	Oil of citronella(Java)
	IS 327: 1991(Reaffirmed Year: 2019)	Oil of lemongrass	European Union	**ISO 3217: 1974**	Oil of lemongrass
	IS 328: 2016(Reaffirmed Year: 2021)	Oil of eucalyptus	European Union	**ISO 3065: 2021**	Oil of Eucalyptus
	IS 15740: 2007 (Reaffirmed Year: 2017)	Oil of rose	European Union	**ISO 9842: 2003**	Oil of rose
	-	-	Australia	**AS ISO 3518: 2022**	Oil of sandalwood
	-	-	Australia	**AS ISO 3848: 2016**	Oil of citronella(Java)
	-	-	Australia	**AS ISO 3217: 1974**	Oil of lemongrass
	-	-	Australia	**AS ISO 3065: 2021**	Oil of eucalyptus
	-	-	Australia	**AS ISO 9842: 2003**	Oil of rose
	-	-	Middle East	**GSO ISO 3518: 2012**	Oil of sandalwood

Domestic Standards			International Standards		
Stage	**Code**	**Aspects Covered**	**Region/ Country**	**Standard**	**Aspects Covered**
	-	-	Middle East	**GSO ISO 3848: 2021**	Oil of citronella(Java)
	-	-	Middle East	**GSO ISO 3217: 2015**	Oil of lemongrass
	-	-	Middle East	**GSO ISO 3065: 2015**	Oil of eucalyptus
	-	-	Middle East	**GSO ISO 9842: 2009**	Oil of rose

6.1. Table 1: (Domestic & International General Standards)

Stage	Code	Scope
General	**IS 6597: 2001 (Reaffirmed Year: 2017)** - Glossary of terms relating to fragrance and flavour industry	▪ Defines the terms relating to fragrance and flavour industries comprising of natural ingredients (essential oils, oleoresin, balsams, resinoids, absolutes) and synthetic aroma chemicals for producing finished fragrance and flavours composition
	ISO 21625: 2020 - Vocabulary of terms used in manufacturing bamboo and bamboo related products	▪ Vocabulary of terms used in manufacturing bamboo and bamboo related products
	AS ISO 21625: 2020 - Vocabulary of terms used in manufacturing bamboo and bamboo related products	▪ Vocabulary of terms used in manufacturing bamboo and bamboo related products

6.2. Table 2: (Domestic Standards for Raw Materials)

Stage	Code	Scope
Raw material	**IS 3349: 1989 (Reaffirmed Year: 2020)** - Perfumery materials - Resinoid benzoin, pure	▪ Prescribes the requirements and the methods of sampling and test for resinoid benzoin, pure
	IS 6874: 2008 (Reaffirmed Year: 2019) - Methods of test of bamboo	▪ Method of tests for determining the following physical and mechanical properties of round bamboo
	IS 326: Part 1: 2018 - Methods of sampling and test for natural and synthetic perfumery material **IS 326: Part 2: 1918 (Reaffirmed Year: 2017)** - Methods of sampling and test for natural and synthetic perfumery material **IS 326: Part 7: 2006 (Reaffirmed Year: 2017)** - Methods of sampling and test for natural and synthetic perfumery material **IS 326: Part 8: 2005 (Reaffirmed Year: 2017)** - Methods of sampling and test for natural and synthetic perfumery material	▪ Prescribes methods of sampling for natural and synthetic perfumery materials, for the purpose of determining the organoleptic, physical and chemical characteristics ▪ Prescribes the special tests for the detection of common impurities and adulterants in natural and synthetic perfumery materials ▪ Determination of acid values ▪ Determination of ester value issued by the International Organization for Standardization (ISO) was adopted by the Bureau of Indian Standards on the recommendations of the Natural and Synthetic Fragrance Materials Sectional Committee and approval of the Petroleum, Coal and Related Products Division Council

6.3. Table 3.a: (Domestic Standards for Processing)

Stage	Code	Scope
Processing	**IS 329: 2004 (Reaffirmed Year: 2020)** - Oil of sandalwood	▪ Requirements and the methods of sampling and test for oil of sandalwood
	IS 512: 1988 (Reaffirmed Year: 2019) - Oil of citronella (Java)	▪ Requirements and the method of sampling and test for oil of citronella (Java)
	IS 327: 1991(Reaffirmed Year: 2019) - Oil of lemongrass	▪ Requirements and the methods of test for the material commercially known as the oil of lemon grass or the East Indian Oil of lemongrass. The material is largely used in the extraction of citral, the chief constituent of the oil and the starting material for the manufacture of important ion-ones. It is also employed in the preparation of lemon-like perfumes, germicides, etc.
	IS 328: 2016(Reaffirmed Year: 2021) - Oil of eucalyptus	▪ Requirements and the methods of sampling and test for oil of Eucalyptus globulus
	IS 15740: 2007 (Reaffirmed Year: 2017) - Oil of rose	▪ Requirements and the methods of sampling and test for oil of rose

6.4. Table 3.b: (International Standards for Processing)

Stage	Code	Scope
Processing	**ISO 3518: 2022** - Oil of sandalwood	▪ Specifies certain characteristics of the oil of sandalwood in order to facilitate assessment of its quality
	ISO 3848: 2016 -Oil of citronella	▪ Specifies certain characteristics of the oil of citronella in order to facilitate assessment of its quality
	ISO 3217: 1974 - Oil of lemongrass	▪ Specifies certain characteristics of the oil of lemongrass in order to facilitate assessment of its quality
	ISO 3065: 2021 - Oil of eucalyptus	▪ Specifies certain characteristics of the oil of eucalyptus in order to facilitate assessment of its quality
	ISO 9842: 2003 - Oil Of rose	▪ Specifies certain characteristics of the oil of rose in order to facilitate assessment of its quality
	AS ISO 3518: 2022 - Oil of sandalwood	▪ Specifies certain characteristics of the oil of sandalwood in order to facilitate assessment of its quality
	AS ISO 3848: 2016 - Oil of citronella	▪ Specifies certain characteristics of the oil of citronella in order to facilitate assessment of its quality
	AS ISO 3217: 1974 - Oil of lemongrass	▪ Specifies certain characteristics of the oil of lemongrass in order to facilitate assessment of its quality
	AS ISO 3065: 2021 - Oil of eucalyptus	▪ Specifies certain characteristics of the oil of eucalyptus in order to facilitate assessment of its quality

Stage	Code	Scope
	AS ISO 9842: 2003 - Oil Of rose	▪ Specifies certain characteristics of the oil of rose in order to facilitate assessment of its quality
	GSO ISO 3518: 2022 - Oil of sandalwood	▪ Specifies certain characteristics of the oil of sandalwood in order to facilitate assessment of its quality
	GSO ISO 3848: 2021 - Oil of citronella	▪ Specifies certain characteristics of the oil of citronella in order to facilitate assessment of its quality
	GSO ISO 3217: 2015 - Oil of lemongrass	▪ Specifies certain characteristics of the oil of lemongrass in order to facilitate assessment of its quality
	GSO ISO 3065: 2015 - Oil of eucalyptus	▪ Specifies certain characteristics of the oil of eucalyptus in order to facilitate assessment of its quality
	GSO ISO 9842: 2009 - Oil of rose	▪ Specifies certain characteristics of the oil of rose in order to facilitate assessment of its quality

Note: The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

REFERENCES

1. Product information- http://odopup.in/en/article/kannauj
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3. ISO Standards: https://www.iso.org/obp/ui#search
4. International Standards:
    a. EU: https://www.en-standard.eu/
    b. USA: https://www.en-standard.eu/astm-standards/ , https://www.astm.org/
    c. AUS: https://www.standards.org.au/
    d. GULF: https://www.gso.org.sa/store/
5. NABL accredited laboratories: https://nabl-india.org/ , https://parakh.ncog.gov.in
6. Domestic Legislation/Regulation: https://www.indiacode.nic.in/handle/123456789/7192?view_type=browse&sam_handle=123456789/2485
7. International Legislation/Regulation:
    a. EU: https://www.legislation.gov.uk/eur/2006/2023/contents
    b. USA: https://www.ecfr.gov/current/title-27/chapter-I/subchapter-A/part-21/subpart-D/section-21.72
    c. AUS: https://www.compliancegate.com
    d. GULF (GCC): https://sabersaudi.co/services/saso-product-coc/saso-technical-regulation

SANITARY FITTINGS, MATHURA

1. INTRODUCTION

Mathura district is famous for the manufacture of sanitary fittings such as water taps, flush cocks, taper cocks and various other sanitary fixtures and components. Apart from being a religious city, Mathura is also well-known centre for manufacture and supply of sanitary fittings throughout Uttar Pradesh. The units in Mathura are known for their ingenuity and wide. There are several micro and small units involved in production of taps in the district.

2. RAW MATERIALS

To make a tap and cock, the raw materials used are:

- Cast brass
- Copper alloy
- Rubber seal/O- rings

3. MANUFACTURING PROCESS (Water Taps, Taper Cock, Push Lift Cock)

The manufacturing process for faucets has become highly automated, with computers controlling most of the machines. Productivity and efficiency have thus improved over the years. The basic process consists of forming the main body of the faucet (some-times including the spout if no swivel is needed), applying a finish and then assembling the various components, followed by inspection and packaging.

3.1. Forming

a. There are two methods used to make the faucet bodies. Most manufacturers use a machining process to shape the body into the required size and dimensions. This involves first cutting the bars into short slugs and automatically feeding them into a computerized, numerically controlled machining centre of multi-spindle and multi-axis design. This machine performs taming, milling, and drilling operations. It typically takes about one minute to make a part.

b. Larger faucets may require numerous machining operations.

For instance, over 32 machining operations are required for some kitchen faucet bodies using a rotary machining centre. With the proper machine, it can take as little as 14 seconds to make a part. Some parts, such as cast spouts for kitchen faucets, are also machined in a separate operation before assembly.

c. Some faucet manufacturers use hot forging instead of machining since this method can produce a near-net shape in about three seconds with little waste. Forging is the process of shaping metals utilizing localised compressive forces. In hot forging, heated metal is forced into a die that is almost the same shape as the faucet body. The pressure is slowly increased over the course of several seconds to make sure the die is completely filled with metal. Only minor machining is required to produce the exact dimensions.

3.2. Finishing

a. After machining, the parts are ready for the finishing process. Those components that come into contact with water may first require a special surface treatment to remove any remaining lead. This involves a leaching process that eliminates lead molecules from the brass surface. The conventional finish is chrome since this material is most resistant against corrosion. First a base coating of electroplated nickel is applied, followed by a thin coating of electroplated chromium. The chrome layer is deposited from a plating bath containing certain additives that improve corrosion resistance.

b. If brass plating is used, a clear polymer coating is applied to improve durability. For white and other coloured finishes, a similar polymer or epoxy plastic with colour added is sprayed onto the faucet in an electrically charged environment. Both coatings then are heat cured.

c. To achieve a polished brass look, Physical Vapour Deposition is used, which applies the metal coating in a vacuum chamber. This chamber has four components: a vacuum pump to provide a controlled environment free of contaminants; a tank that emits several types of gases; a target rod that acts as the metal source and racks to hold the faucet parts. The target rod is made of a corrosion-resistant material such as zirconium.

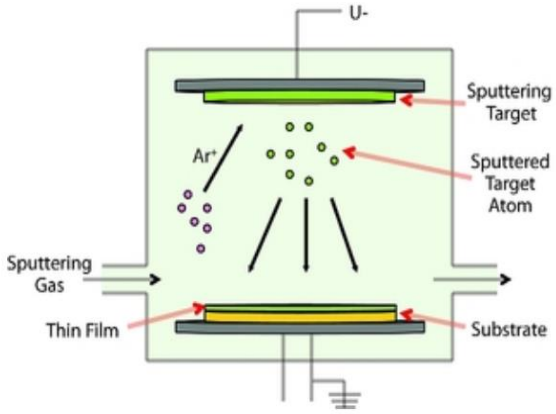

Figure 1: Physical Vapour Deposition

d. An electric arc heats the target to vaporize the material, then strikes the surface of the faucet at high speed and reacts with the mixture of gases to provide colour and corrosion resistance. As the target material combines with these gases, it adheres to the faucet part, creating a bond that is virtually indestructible. Some manufacturers use a spiral coil around the target to provide a uniform distribution of the coating.

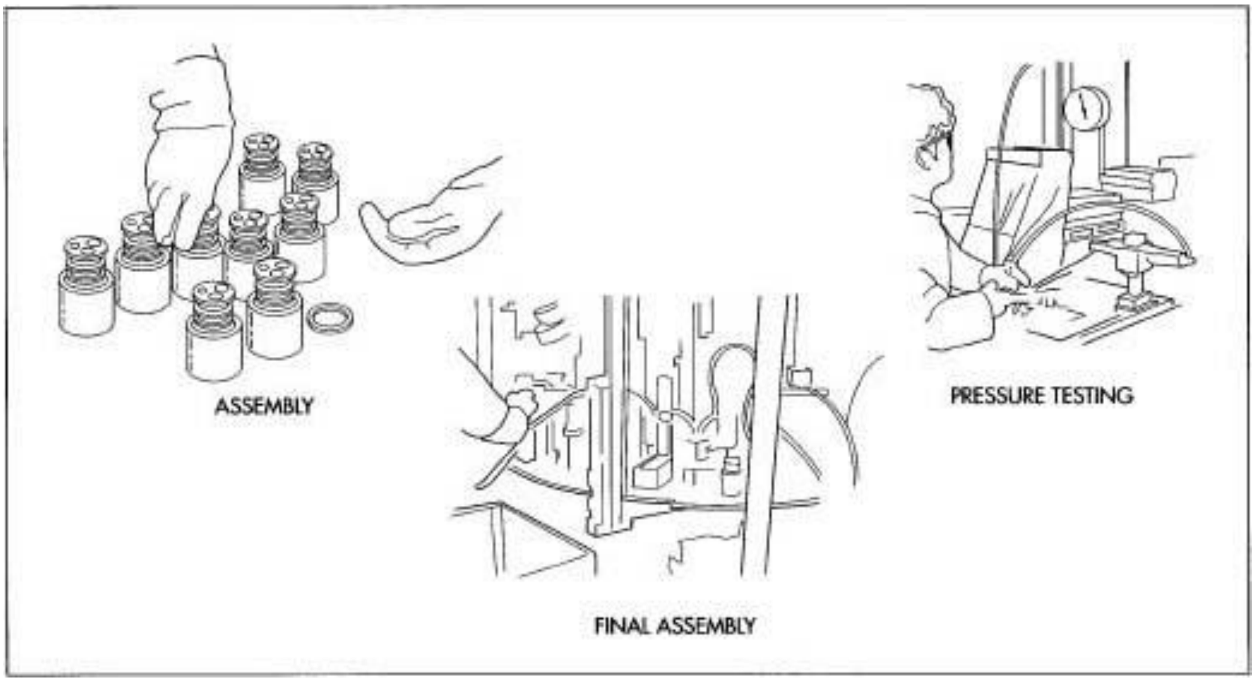

Figure 2: Pressure testing of finished products

3.3. Assembly

a. After plating, the parts are stored in bins until assembly. Assembly can involve both manual and automated processes. For some faucets, pre-lubricated rubber seals or O-rings are installed by hand.

b. Finally, faucets and other components are sent for the final assembly. This process takes place on rotary assembly machines which are precisely controlled. The sprout, if separate, is first installed, followed by the ceramic cartridge. This cartridge is screwed in place with a brass using a pneumatic gun, and then the handle is attached by hand. Sometimes the copper tubes are installed before assembly. After assembly, the faucets are packaged in boxes along with any other components that are needed for final installation.

4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- Checking under cold-water pressure or under air pressure the water tightness of bonnet assembly on seat, tap and valve upstream, taps and valve
- There shall be no leakage of water or escape of air bubbles respectively, through the walls of the body, the bonnet and the diverter assembly
- Finished product should not deform during testing
- Plating to be uniform, inspect for chipped coatings

5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations (domestic) and international (some key regions/countries) are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

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5.1. Domestic

S. No.	Country	Regulation	Regulatory body	Remarks
1	India	NA	NA	NA

5.2. International

S. No.	Country	Regulation	Regulatory body	Remarks
1	European Union	**a.** Restrictions on Hazardous Substances (RoHS) Regulation(EC) 863/2015 **b.** Regulation (EU) 2021/854 of 27 May 2021	**a.** European Commission	**a.** Compliance on not having more than the prescribed limit for Lead, Chromium and Hexavalent chromium **b.** Anti-Dumping legislations on imports of stainless steel cold rolled flats originating in India and Indonesia
2	United States of America	**a.** California Proposition 65, Act of 1986. **b.** Restrictions on Hazardous Substances (RoHS) Directive	**a.** California Government **b.** Respective State Governments having RoHS regulations in place	**a.** Any manufacturing unit having more than ten employees are required to label products if they contain carcinogenic products **b.** Compliance on not having more than the prescribed limit for Lead, Chromium and Hexavalent chromium
3	Middle East	**a.** Saudi Standard, Metrology and Quality Organisation(SASO) Regulations	**a.** Saudi Standard, Metrology and Quality Organisation (SASO)	**a.** General product safety regulation **b.** Labelling Requirement

6. VOLUNTARY STANDARDS

Domestic Standards			International Standards		
Stage	**Code**	**Aspects Covered**	**Region/ Country**	**Standard**	**Aspects Covered**
Raw Materials (Refer Table 1)	**IS 292: 1983 (Reaffirmed Year: 2021)**	Leaded brass ingots and castings	-	-	-
	IS 9975: Part 1: 1981 (Reaffirmed Year: 2019)	O Rings – Dimensions, material selection and quality acceptance criteria, tolerances and design criteria for standard applications	-	-	-
	IS 319: 2007 (Reaffirmed Year: 2022)	Free cutting brass bars, rods and section	-	-	-
	IS 410: 1997 (Reaffirmed Year: 2021)	Cold rolled brass sheet, strip and foil	-	-	-
	IS 1264: 1997 (Reaffirmed Year: 2021)	Brass gravity die castings (ingots and castings)	-	-	-
	IS 6912: 2005 (Reaffirmed Year: 2021)	Copper and copper alloy forging stock and forgings	-	-	-
	IS 318: 1981 (Reaffirmed Year: 2021)	Leaded tin bronze ingots and castings	-	-	-

Domestic Standards			International Standards		
Stage	**Code**	**Aspects Covered**	**Region/ Country**	**Standard**	**Aspects Covered**
	IS 319: 2007 (Reaffirmed Year: 2022)	Free cutting brass bars, rods and section	-	-	-
Manufacturing/ Processing (Refer Table 2a. & 2.b)	**IS 9975: Part 1: 1981 (Reaffirmed Year: 2019)**	Dimensional requirements 'O' Rings.	European Union	**BS EN 232: 2012**	Baths. Connecting dimensions
	IS 781: 1984 (Reaffirmed Year: 2020)	Construction and workmanship requirements for cast copper alloy screw down bib taps and stop valves for water services	European Union	**DIN EN 200: 2008**	Sanitary Tap ware - Single Taps And Combination Taps For Water Supply Systems Of Type 1 And Type 2 - General Technical Specification
	IS 1711: 1984 (Reaffirmed Year: 2020)	Manufacturing requirements for self-closing taps for water supply purposes	European Union	**DIN EN 1111: 2017**	Sanitary Tap ware - Thermostatic Mixing Valves (PN 10) - General Technical Specification
	IS 1795: 1982 (Reaffirmed Year: 2020)	Construction requirements for pillar taps.	European Union	**DIN EN 817: 2008**	Sanitary Tap ware - Mechanical Mixing Valves (PN 10) - General Technical Specifications
	IS 8931: 2021	Dimensional and constructional requirements of copper alloy fancy single taps, combination tap assembly and stop	USA	**ICS 91.140.70**	Sanitary installation - Including bidets, kitchen sinks, baths, waste chutes, etc.
	IS 9763: 2000 (Reaffirmed Year: 2020)	Plastics Bib taps, pillar taps, angle valves and stop valves for hot and cold water services	-	-	-
	IS 14845: 2000 (Reaffirmed Year: 2020)	Cast iron air relief valves for water works	-	-	-
	IS 13114: 1991 (Reaffirmed Year: 2017)	Forged brass gate, globe and check valves for water works purposes	-	-	-
	IS 9975: Part 1: 1981 (Reaffirmed Year: 2019)	Specifications for O rings	-	-	-
Finishing (Refer Table 3)	**IS 781: 1984 (Reaffirmed Year: 2020)**	Finishing and sampling requirements for cast copper alloy screw down bib taps and stop valves for water services	-	-	-
	IS 1711: 1984 (Reaffirmed Year: 2020)	Finishing requirements quality tests for self-closing taps for water supply	-	-	-

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/ Country	Standard	Aspects Covered
	IS 1795: 1982 (Reaffirmed Year: 2020)	Finishing, sampling and testing requirements for pillar taps	-	-	-

6.1. Table 1: (Domestic Standards for Raw Materials)

Stage	Code	Scope
Raw Material	**IS 292: 1983 (Reaffirmed Year: 2021)** - Leaded brass ingots and castings	▪ Covers the requirements for two grades of leaded brass ingots and casting designated as LCB 1 and LCB 2
	IS 9975: Part 1: 1981 (Reaffirmed Year: 2019) - Specification for \`O\` Rings - Material selection and quality acceptance criteria	▪ Covers the material, selection requirements, and quality acceptance criteria for 'O'-Rings for reciprocating and static purposes
	IS 319: 2007 (Reaffirmed Year: 2022) - Free cutting brass bars, rods and section	▪ Covers the requirements for flee cutting brass bars, rods and sections having a minimum cross sectional dimension over 6 mm suitable for high speed screw cutting and turning work
	IS 410: 1997 (Reaffirmed Year: 2021) - Cold rolled brass sheet, strip and foil	▪ Covers the requirements of three alloys of cold rolled brass sheet strip and foil required for engineering and general purposes and designated as CuZn30, CuZn37 and CuZn40
	IS 1264: 1997 (Reaffirmed Year: 2021) - Brass gravity die castings (ingots and castings)	▪ Covers the requirements for two grades, of brass gravity die castings DCB 1 and DCB2
	IS 6912: 2005 (Reaffirmed Year: 2021) - Copper and copper alloy forging stock and forgings	▪ Specifies requirements for chemical composition, mechanical properties and other characteristics for wrought or cast copper and alloy forging stock and for forgings of these materials
	IS 318: 1981 (Reaffirmed Year: 2021) - Leaded tin bronze ingots and castings	▪ Covers the requirements for six grades of leaded tin bronze ingots and castings
	IS 319: 2007 (Reaffirmed Year: 2022) - Free cutting brass bars, rods and section	▪ Covers the requirements for flee cutting brass bars, rods and sections having a minimum cross sectional dimension over 6 mm suitable for high speed screw cutting and turning work

6.2. Table 2.a: (Domestic Standards for Processing)

Stage	Code	Scope
Processing	**IS 9975: Part 1: 1981 (Reaffirmed Year: 2019)**- Specification for \`O\` Rings	▪ Quality Tests requirements on 'O' rings ▪ Workmanship and Finishing requirements on 'O' rings
	IS 781: 1984 (Reaffirmed Year: 2020) - Cast copper alloy screw down bib taps and stop valves for water services	▪ Material, selection requirements, and quality acceptance criteria for '0'~Rings ▪ Requirements of copper alloy screw down bib taps and stop valves suitable for cold non-shock working pressure

Stage	Code	Scope
	IS 1711: 1984 (Reaffirmed Year: 2020) - Self-closing taps for water supply purposes	▪ Standard lays down requirements for self-closing taps with or without stuffing box ▪ Manufacturing material requirements of components
	IS 1795: 1982 (Reaffirmed Year: 2020) - Pillar taps for water supply purposes	▪ Requirements regarding material, manufacture and construction, finish and testing of pillar taps
	IS 8931:2021 - Copper alloy fancy single taps, combination tap assembly and stop valves for water services	▪ Requirements regarding manufacturing, workmanship, and construction of chromium plated copper alloy non-rising spindle type**:** fancy single taps, combination tap assembly and valves
	IS 9763: 2000 (Reaffirmed Year: 2020) - Plastics bib taps, pillar taps, angle valves and stop valves for hot and cold water services	▪ Requirements regarding material Dimensions and construction, of plastic bib taps, pillar taps, stop valves and angle valves for hot and cold water services
	IS 14845: 2000 (Reaffirmed Year: 2020) - Resilient seated cast iron air relief valves for water works purposes	▪ Requirement of single air valve (small and large orifice), double air valves (small and large orifice with or without integral isolating valve) and kinetic air valves with or without separate isolating sluice valve for use on water mains
	IS 13114: 1991 (Reaffirmed Year: 2017) - Forged brass gate, globe and check valves for water works purposes	▪ Requirements for forged brass gate, globe and check valves suitable for working temperatures up to 45°C and non-shock maximum hydraulic working pressure of 2MPa for water works purposes

6.3. Table 2.b: (International Standards for Processing)

Stage	Code	Scope
Processing	**BS EN 232: 2012** - Baths-Connecting dimensions	▪ Specifies requirements for the connecting dimensions of baths, regardless of the material used for their manufacture. This European Standard applies to baths used for domestic purposes and complements the standards for baths made from different materials
	DIN EN 200: 2008 - Sanitary tap ware - Single taps and combination taps for water supply systems of type 1 and type 2 - general technical specification	▪ Specifies - the field of application for pillar taps, bib taps, single and multi-hole combination taps for use in supply systems of type 1 and type 2; -the dimensional, tightness, pressure resistance, hydraulic, mechanical strength, endurance and acoustic characteristics of nominal size 1/2 and 3/4 single taps and combination taps as well as the test methods to verify the characteristics
	DIN EN 1111: 2017 - Sanitary tap ware - Thermostatic mixing valves (PN 10) - General technical specification	▪ Specifies construction, performance and material requirements for high pressure supplied thermostatic mixing valves and includes test methods for the verification of mixed water temperature performance at the PoU below 45°C

Stage	Code	Scope
	DIN EN 817: 2008 - Sanitary tap ware - Mechanical mixing valves (PN 10) - General technical specifications	▪ Specifies the dimensional, leak tightness, mechanical and hydraulic performance, mechanical endurance and acoustic characteristics of mechanical mixing valves as well as the procedures for verification of these characteristics
	ICS 91.140.70 - Sanitary installation - Including bidets, kitchen sinks, baths, waste chutes, etc.	▪ Sanitary installation - Including bidets, kitchen sinks, baths, waste chutes, etc.

6.4. Table 3: (Domestic Standards for Finishing)

Stage	Code	Scope
Finishing	**IS 781: 1984 (Reaffirmed Year: 2020)** - Cast copper alloy screw down bib taps and stop valves for water services	▪ Finishing Requirements for Bib taps and Stop valves. ▪ Sampling & Marking requirements
	IS 1711: 1984 (Reaffirmed Year: 2020) - Self-closing taps for water supply purposes	▪ Finishing requirements for self-closing taps for water supply purposes
	IS 1795: 1982 (Reaffirmed Year: 2020) - Pillar taps for water supply purposes	▪ Finishing requirements for Pillar taps, nickel-chromium plated and thickness of coating ▪ Sampling Procedure and Criteria ▪ Marking Requirements for Pillar Taps

Note: The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

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8. GULF: https://www.gso.org.sa/store/
9. NABL accredited laboratories: https://nabl-india.org/, https://parakh.ncog.gov.in
10. Domestic Legislation/Regulation: https://www.indiacode.nic.in/handle/123456789/7192?view_type=browse&sam_handle=123456789/2485
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GOLDSMITHY, BAREILLY

1. INTRODUCTION

Gold is a malleable, ductile, rare and the only solid metallic element with yellow colour. It may easily be melted, fused, and cast without the problems of oxides and gas that are problematic with other metals such as bronzes, for example. It is fairly easy to "pressure weld", wherein, similarly to clay, two small pieces may be pounded together to make one larger piece. Gold is classified as a noble metal—because it does not react with most elements. It is usually found in its native form, lasting indefinitely without oxidization and tarnishing.

Gold has been worked by humans in all cultures where the metal is available, either indigenously or imported and the history of these activities is extensive. The art of working with gold is called Gold smithy.

A goldsmith is a metalworker who specializes in working with gold and other precious metals. Nowadays, they mainly specialize in jewellery-making but historically, goldsmiths have also made silverware, platters, goblets, decorative & serviceable utensils and ceremonial or religious items.

Many universities and junior colleges also offer goldsmithing, silversmithing, and metal arts fabrication as a part of their fine arts curriculum. Gold is one of the finest metals used for jewellery as it is one of the most malleable and ductile metals, making it easy to work with. Gold is also resistant to most acids. Pure gold will not tarnish, however 14k, 16k, 18k will, however at a rate much less than Silver and over a relatively long time period. Gold is also less allergenic than silver making it one of the most desirable metals for jewellery making.

The Mandatory Hallmarking of Gold has been rolled out in 256 districts where there is at least one Assaying Centre in the country. Uttar Pradesh has made hallmarking of gold mandatory and has been implemented in 19 districts of Uttar Pradesh including Bareilly. This aims to organize the gold industry which is at present a largely unorganized sector.

The enterprises which have an annual turnover of less than 40 Lakhs, however, are exempt from this.

2. RAW MATERIALS

Gold purity is expressed in “Carats”. Pure Gold is extremely soft and to increase its workability, it is usually alloyed with copper, silver, zinc, nickel, or palladium to produce a harder material for the jewellery trade. The primary raw materials are:

- Gold
- Copper, silver, zinc etc. for alloys
- Gold solder

3. MANUFACTURING PROCESS

Jewellery manufacturing is a very complex procedure, often involving preparing separate pieces and then assembling them.

3.1. Melting the Metal

Gold is heated in a crucible usually made of graphite or clay to 1064 °C. Flux (a mixture of borax and sodium carbonate) is added to remove all impurities from the gold. This molten gold after extraction of the impurities is moulded into ingots.

Figure 1: Melting gold in a crucible

3.2. Making Strips

The gold ingots are rolled under heavy wheels to obtain wires and sheets depending on the design.

3.3. Making the Pieces

The artisans cut pieces from the plates and wires and fashion the different pieces. A jewellery product like a necklace or bangle can comprise of many different pieces. All these pieces are crafted individually by artisans. These pieces are set on a wax plate to exactly replicate the design and soldered together to form a solid piece. The residue left behind on the gold is cleaned using chemicals.

Figure 2: Setting on a wax plate

3.4. Polishing

The filigree and smaller parts are hand polished while the larger pieces are polished using the polishing wheels.

3.5. Cleaning

The product is now placed in an ultrasonic shaker machine which emits ultrasonic waves to shake any dirt adhering to inaccessible crevices.

3.6. Setting of Stones

If the design further includes precious stones, they are set after the cleaning process. The jewellery piece is mounted on a lac base atop a wooden dumbbell. The artisan carefully places the stones in the cavities and the stones are secured using the usually seen prong method.

Figure 3: Stone Setting

4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- The artefact should not have scratches or dents
- The jewellery should be thoroughly cleaned and not contain any residue from the manufacturing processes

5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations (domestic) and international (some key regions/countries) are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organisations (https://www.fieo.org/)
- Directorate General of Foreign Trade (https://www.dgft.gov.in/CP/)
- Export Promotion Councils (https://commerce.gov.in/useful-links/export-promotion-councils/)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' (FTAs) with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (https://commerce.gov.in/international-trade/trade-agreements/).

5.1. Domestic

S. No.	Country	Regulation Reference	Regulatory Body	Remarks
1	India	BIS Hallmarking Scheme	Bureau of Indian Standards (BIS)	**a.** Hallmarking is the accurate determination and official recording of the proportionate content of precious metal in precious metal articles **b.** IS 1417: 2016:2016 permits gold jewellery/artefacts of 14 carats, 18 carats and 22 carats to be hallmarked and sold. However, an amendment to the Indian Standard is being issued which would permit Hallmarking of six caratages of gold jewellery/ artefacts, viz. 14, 18, 20, 22, 23 and 24 carats

5.2. International

S. No.	Country	Regulation Reference	Regulatory Body	Remarks
1	European Union	**a.** Conflict Minerals Regulation (2017/821) **b.** Registration, Evaluation, Authorization and Restriction of Chemicals (REACH) regulations, (EC 1907/2006) **c.** Hallmarking of Jewellery	**a.** European Commission **b.** European Chemicals Agency (ECHA)	**a.** Prevention of imports of conflict metals (Tungsten, Tantalum, Tin and Gold) **b.** Hazardous chemicals found in jewellery such as lead, chromium, phthalates, nickel, cadmium etc. **c.** Jewellery to be assayed by the local assayer's office for purity **d.** Prevention of import of jewellery made from endangered animal parts
2	United States of America	**a.** Consumer Product Safety Improvement Act (CSPIA) Act of 2008 **b.** Federal Hazardous Substances Act (FHSA), Act of 1960. **c.** California Proposition 65, Act of 1986. **d.** Children's Product Certificate	**a.** Consumer Product Safety Commission (CPSC) **b.** California State Government	**a.** CPSIA tracking label displaying: ▪ Importer company name ▪ Production location ▪ Production date ▪ Batch number **b.** Hazardous Chemicals **c.** Importers liability to draft and issue a Children's Product Certificate, conforming adherence to mandatory standards(ASTM F963, ASTM F2923)
3	Middle East	**a.** Dubai Central Laboratories Department (DCL)	**a.** Ministry of Economy In The UAE	**a.** Bareeq certification **b.** The purity of gold jewellery sold in Dubai is ensured by the Dubai Central Laboratories Department (DCL)

6. VOLUNTARY STANDARDS

Domestic Standards			International Standards		
Stage	**Code**	**Stage**	**Region/ Country**	**Standard**	**Aspects Covered**
General (Refer Table 1)	**IS 15892: 2011 (Reaffirmed Year: 2016)**	Ring-Sizes - Definition measurement & designation mentioned	USA	**ISO 8653: 2016**	Ring-Sizes - definition measurement & designation mentioned
	IS 17278: 2017	Refined gold and silver bars for good delivery — Specification	-	-	-
	IS 15820: 2009 (Reaffirmed Year: 2019)	General requirements for competence of assaying and hallmarking centre	-	-	-

Domestic Standards			International Standards		
Stage	**Code**	**Stage**	**Region/ Country**	**Standard**	**Aspects Covered**
Raw Materials (Refer Table 2.a & 2.b)	**IS 1417: 2016: 2016**	Gold and gold alloys, jewellery/artefacts - Fineness and marking	USA	**ISO 8654: 2018**	Colours of gold alloys related to jewellery
	IS 14741: 1999 (Reaffirmed Year: 2020)	Colours of gold alloys - Definition, range of colours and designation [MTD 10: Precious Metals]	Australia	**CS - 030**	Precious metals for jewellery
	-	-	Australia	**AS 2335**	Requirements for rolled gold and plated jewellery
Manufacturing/ Processing (Refer Table 3.a & 3.b)	**IS 7255: Part 1: 1974 (Reaffirmed Year: 2019)**	Methods of chemical analysis of solders for use in gold ware.	Australia	**AS 2140**	Jewellery - Fineness of precious metal alloys
	IS 7255: Part 2: 1977 (Reaffirmed Year: 2019)	Methods of Chemical Analysis of Solders for Use in Goldware - Part II : Determination of Cadmium and Zinc by Polarographic Method	-	-	-
	IS 1418: 2009 (Reaffirmed Year: 2019)	Method for determination of gold in gold bullion, gold alloys and gold jewellery/artefacts - Cupellation (fire assay)	-	-	-
	IS 2790: 1999 (Reaffirmed Year: 2020)	Guidelines for manufacture of 23, 22, 21, 18, 14 and 9 Carat gold alloys	-	-	-
	IS 3095: 1999 (Reaffirmed Year: 2020)	Requirement of gold solders for use in the manufacture of gold jewellery/artefacts of 23, 22, 21, 18, 14 and 9 carat	-	-	-
Finishing (Refer Table 4)	**IS 7739: Part1: 1975 (Reaffirmed Year: 2022)**	Code of for preparation of metallographic specimens	-	-	-

6.1. Table 1: (Domestic & International General Standards)

Stage	Code	Scope
Raw Material	**IS 15892: 2011 (Reaffirmed Year: 2016)** - Ring-Sizes - Definition measurement & designation mentioned	▪ This standard specifies the method for measuring the size of ring for the purposes of designating ring size to be used in jewellery
	IS 17278: 2017 - Specification for refined gold and silver bars for good delivery	▪ Specifies the technical requirements of refined gold and silver bars for good delivery by refiners ▪ The bars shall be produced by the refiners as per the guidelines laid down by the certifying agency
	IS 15820: 2009 (Reaffirmed Year: 2019) - General requirements for competence of assaying and hallmarking centre	▪ This standard covers general requirements for competence of assaying and hallmarking centres
	ISO 8653: 2016 - Ring-Sizes - Definition measurement & designation mentioned	▪ This International Standard specifies a method to measure the ring-size using a ring stick with defined characteristics, which is mainly used during manufacturing steps, and specifies the designation of the ring-size. NOTE For jeweller-consumer relationships, the finger size is measured with a finger gauge set made up of a ring for each size with the same diameter and tolerance than the ring stick ones

6.2. Table 2.a: (Domestic Standards for Raw Materials)

Stage	Code	Scope
Raw Material	**IS 1417: 2016: 2016** - Gold and gold alloys, jewellery/artefacts fineness and marking	▪ Specifies based on gold content**:** **a.** Grades of fine gold and standard gold used for manufacture of bullion and coins **b.** Grades of gold alloy used in manufacture of jewellery/artefacts ▪ Specifies the guidelines for marking of purity and other details on tested bullion, coins and jewellery/artefacts NOTE — Artefacts cover medallions and utensils
	IS 14741: 1999 (Reaffirmed Year: 2020) - Colours of gold alloys - Definition, range of colours and designation [MTD 10**:** Precious Metals]	▪ Specifies a limited number of colours of gold alloys ▪ It applies to jewellery, and to watch cases and accessories made of gold alloys or watch cases and accessories with gold alloy coverings

6.3. Table 2.b: (International Standards for Raw Materials)

Stage	Code	Scope
Raw Material	**ISO 8654: 2018** - Colours of gold alloys related to Jewellery	▪ Standardization in the field of jewellery (e.g. numbering system, sizes of rings, precious metals colours and coatings, diamonds) and precious metals (e.g. analyses, sampling, impurities)
	CS - 030 - Precious metals for jewellery	▪ Specify standards for precious metals for jewellery
	AS 2335 - Requirements for rolled gold and plated jeweller	▪ Specifies composition and marking requirements for rolled gold and plated jewellery

6.4. Table 3.a: (Domestic Standards for Processing)

Stage	Code	Scope
Processing	**IS 7255: Part 1: 1974 (Reaffirmed Year: 2019)** - Methods of chemical analysis of solders for use in gold ware	▪ This standard covers the methods for determination of gold, silver and copper in solders for use in gold wares as specified in IS 3095: 1999 (Reaffirmed Year: 2020) : 1966
	IS 7255: Part 2: 1977 (Reaffirmed Year: 2019) - Methods of Chemical Analysis of Solders for Use in Goldware - Part II : Determination of Cadmium and Zinc by Polarographic Method	▪ Methods of Chemical Analysis of Solders for Use in Goldware - Part II : Determination of Cadmium and Zinc by Polarographic Method
	IS 1418: 2009 (Reaffirmed Year: 2019) - Method for determination of gold in gold bullion, gold alloys and gold jewellery/artefacts - Cupellation (Fire Assay)	▪ This standard prescribes the cupellation or fire assay method for assaying of gold in gold bullion, gold alloys, and gold jewellery/artefacts covered in IS 1417: 2016
	IS 2790: 1999 (Reaffirmed Year: 2020) - Guidelines for manufacture of 23, 22, 21, 18, 14 and 9 carat gold alloys	▪ This standard covers guidelines for manufacture of 23, 22, 21,18, 14 and 9 carat gold in the form of ingot, bar, plate, sheet, rod, wire and granules
	IS 3095: 1999 (Reaffirmed Year: 2020) - Requirement of gold solders for use in the manufacture of gold jewellery/artefacts of 23, 22, 21, 18, 14 and 9 carat	▪ This standard lays down the requirements for gold solders for use in the manufacture of gold jewellery/artefacts of 23, 22, 21,18, 14 and 9 carat

6.5. Table 3.b: (International Standards for Processing)

Stage	Code	Scope
Processing	**AS 2140** - Jewellery - Fineness of precious metal alloys	▪ This Standard sets out the composition and marking requirements for precious metal articles, including articles of jewellery NOTE: This specification does not apply to articles intended for industrial purposes

6.6. Table 4: (Domestic Standards for Finishing)

Stage	Code	Scope
Finishing	**IS 7739: Part1: 1975 (Reaffirmed Year: 2022)** -Code of for preparation of metallographic specimens	▪ This standard specifies based on gold content: Grades of fine gold and standard gold used for manufacture of bullion and coins, and Grades of gold alloy used in manufacture of jewellery/artefacts ▪ This standard also specifies the guidelines for marking of purity and other details on tested bullion, coins and jewellery/artefacts NOTE — Artefacts cover medallions and utensils

Note: The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

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GULABI MEENAKARI, VARANASI

1. INTRODUCTION

Minakari or Meenakari is the process of painting and colouring the surfaces of metals and ceramic tiles through enamelling. The word Minakari is a compound word composed of the words 'mina' and 'kari'. 'Mina' is a feminine variation of the word 'minu' which means paradise or heaven. 'Kari' means to do or place something onto something else. Together the word Minakari means to place paradise onto an object.

This art originated in Safavid, Iran. It is practiced as an art form and commercially produced mainly in Iran, Afghanistan, Pakistan and India. Minakari art usually involves intricate designs (mainly using geometric shapes and designs) and is applied as a decorative feature to serving dishes, containers, vases, frames, display ornaments and jewellery.

Meenakari or the art of enameling metal for ornamental reasons has been traced back to the Parthian and Sassanid period of Iranian history. However, the meticulous ornamental work seen today can be traced back to Safavid, Iran around the 15th century. The Moghuls introduced it into India and perfected the technique making the design applied on objects more intricate. The craft reached its peak in Iran during the eighteenth and nineteenth century. In the twentieth century, Iranian artisans specializing in meenakari were invited to other regions to assist with training local craftsmen. Varanasi is an important hub for this art.

It is an art of ornamenting metal surface with attractive colours and kundhan stones. In olden days, the meenakari art was done on gold but later on other metals like silver, copper, etc. also found the way. The artisans who practice the art of meenakari are known as meenakars. Meenakari is not just limited to traditional jewellery but expands into more modern products including bowls, ashtrays, key chains, figures of deities and decorative showpieces.

Meenakari is famous in various places in India and Varanasi is one among them. The meenakari in Varanasi is characterized by pink strokes on white enamel hence it is known as Banaras Gulabi Meenakari. The product received the GI tag in the year 2014, which helped enhance its presence in the global market as a unique product.

2. RAW MATERIALS

The raw materials for Gulabi Meenakari are as below:

- Silver Sheet
- Glass materials for preparation of enamel
- Pomegranate juices mixed in water
- Water soluble flux

3. MANUFACTURING PROCESS

The process of manufacturing is as described below:

3.1. Silver Sheet Cutting

Silver sheet is the basic raw material for meenakari art. A portion of the silver sheet is cut according to the required size of the article being manufactured. The sheet is then placed over an embossing tool and hammered to obtain the raised impression. This sheet is heated to make it more pliable and easy to bend. Once done, the left-over silver foil around the raised design is carefully cut away. Multiple parts are created separately.

3.2. Joining

The multiple parts are joined together using a silver wire and reheated to fuse the two parts together. The entire joinery process continues till all pieces are joined together.

Figure 1: Joining the parts

3.3. Filing

The entire artefact is filed carefully to remove all rough edges and give a smooth, uniform and neat finish. The different parts are then mounted onto a pedestal.

3.4. Enamelling

The enamelling process starts with grinding different coloured glass in a pestle with pomegranate seeds and water. This gives a colour which is bright and glossy. The meenakar or the artisan who paints the handicraft then applies the colour to the artefacts.

Figure 2: Applying colour

3.5. Heating in the furnace

The enamelled craft is heated in the furnace where the colours fuse and harden to become one with the surface for long lasting glossy surface. The heat of the furnace is set between 750ºC to 850ºC. The kundan stones are added to embellish the artwork. It is important to ensure that the hardest colour is used first and fired. Then the metal is cooled to start with the detailing. The preheated craft is painted with the unique colour combination of pink on white, which is again heated in the furnace to get the characteristic effect on the product.

4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- Check for uniformity in product or any other visual defects such as cracks, chipping etc.
- The welding joints to be inspected for clean finish

5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations (domestic) and international (some key regions/countries) are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organisations (https://www.fieo.org/)
- Directorate General of Foreign Trade (https://www.dgft.gov.in/CP/)
- Export Promotion Councils (https://commerce.gov.in/useful-links/export-promotion-councils/)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (https://commerce.gov.in/international-trade/trade-agreements/).

5.1. Domestic

S. No.	Country	Regulation Reference	Regulatory Body	Regulation
1	India	NA	NA	NA

5.2. International

S. No.	Country	Regulation Reference	Regulatory Body	Regulation
1	European Union	a. Registration, Evaluation, Authorization and Restriction of Chemicals (REACH) Regulations, (EC 1907/2006)	a. European Commission b. European Chemical Agency (ECHA)	a. Heavy chemicals
2	United States of America	a. California Proposition 65 (1986) b. Consumer Product Safety Act (CPSA)	a. Consumer Product Safety Commission (CPSC)	a. Heavy Chemical - Children jewellery b. Hazardous Chemicals

S. No.	Country	Regulation Reference	Regulatory Body	Regulation
			b. California State	
3	Australia	a. Country of Origin Labelling	a. Australian Consumer Law	a. Country of labelling
4	Middle East	a. Saudi Standards, Metrology and Quality Organization (SASO) Regulations b. SASO Certificate of Conformity	a. Safety, Saudi Standards, Metrology and Quality Organization (SASO)	a. General product safety regulations b. Labelling of product

6. VOLUNTARY STANDARDS

Domestic Standards			International Standards		
Stage	**Code**	**Aspects Covered**	**Region/ Country**	**Standard**	**Aspects Covered**
General (Refer Table 1)	-	-	USA	**ASTM F2999-19**	Standard Consumer Safety Specification for Adult Jewellery
	-	-	USA	**ASTM F2923-20**	Standard Specification for Consumer Product Safety for Children's Jewellery
Raw Materials (Refer Table 2.a & 2.b)	**IS 2279: 2018 (Reaffirmed Year: 2019)**	Fine silver bar, sheet, wire, granules and token		**ISO 9227: 2017**	Corrosion tests in artificial atmospheres — Salt spray tests
	IS 2932: Part 1: 2013 (Reaffirmed Year: 2018)	Enamel, synthetic, exterior: (a) Undercoating (b) Finishing - Specification: Part 1 for Domestic and decorative applications	Middle East	**GSO ISO 10308: 2013**	Metallic coatings- Review of porosity tests
Manufacturing/ Processing (Refer Table 3a. & 3.b)	**IS 964: 1956 (Reaffirmed Year: 2019)**	Methods for chemical analysis of silver solder	European Union	**BS ISO 15096: 2020**	Jewellery and precious metals. Determination of high purity silver. Difference method using ICP-OES
	IS 2113: 2014 (Reaffirmed Year: 2019)	Enamel, Synthetic, Exterior : (a) Undercoating (b) Finishing - Specification : Part 1 for Domestic and Decorative Applications	European Union	**BS EN 1904: 2000**	Precious metals. The fineness's of solders used with precious metal jewellery alloys
	IS 1067: 1981 (Reaffirmed Year: 2021)	Electroplated coatings of silver for decorative and protective purposes	European Union	**BS EN ISO 11427: 2016**	Jewellery. Determination of silver in silver jewellery alloys. Volumetric

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/ Country	Standard	Aspects Covered
					(potentiometric) method using potassium bromide
	-	-	Australia	**AS 1856: 2004**	Electroplated coatings - Silver
	-	-	Australia	**AS 3895.1-1991**	Methods for the analysis of copper, lead, zinc, gold and silver ores, Part 1: Determination of gold (Fire assay - Flame AAS method)
	-	-	Australia	**ISO 13756: 2015**	Jewellery - Determination of silver in silver jewellery alloys - Volumetric (potentiometric) method using sodium chloride or potassium chloride
Finishing (Refer Table 4.a & 4.b)	**IS 1417: 2016**	Gold and gold alloys, jewellery/artefacts - Fineness and marking	European Union	**BS EN ISO 9202: 2019**	Jewellery and precious metals. Fineness of precious metal alloys
	IS 2112: 2014 (Reaffirmed Year: 2019)	Silver and silver alloys, jewellery/artefacts - Fineness and marking - Specification (Third Revision)	USA	**ASTM D7786-13**	Standard test method for determining enamel holdout
	IS 101: Part 4: Section 2: 2021	Methods of sampling and test for paints, varnishes and related products	USA	**ASTM 3359**	Standard test methods for rating adhesion by tape test
	IS 101: Part 4: Section 4: 2020	Methods of sampling and test for paints, varnishes and related products	USA	**ASTM D 6659**	Standard practice for xenon-arc exposures of paint and related coatings
	IS 101: Part 5: Section 1: 1988 (Reaffirmed Year: 2019)	Methods of sampling and test for paints, varnishes and related products Part 5 Mechanical test on paint films Sec 1 Hardness tests	-	-	-
	IS 101: Part 8: Section 2: 1990 (Reaffirmed Year: 2017)	Methods of sampling and test for paints, varnishes and related products Part 8 Tests for pigments and other solids Sec 2 Pigments and non-volatile matter	-	-	-

6.1. Table 1: (Domestic & International General Standards)

Stage	Code	Scope
General	**ASTM F2999-19** - Standard consumer safety specification for adult jewellery	▪ This specification establishes requirements and test methods for specified elements and for certain mechanical hazards in adult jewellery ▪ It does not purport to cover every conceivable hazard of adult jewellery ▪ It does not cover product performance or quality, except as related to safety ▪ This specification has no requirements for those aspects of adult jewellery that present an inherent and recognized hazard as part of the function of jewellery
	ASTM F2923-20 - Standard specification for consumer product safety for children's jewellery	▪ This specification establishes requirements and test methods for specified elements and certain mechanical hazards in children's jewellery ▪ It also includes recommendations for age labelling and warnings, and guidelines on identifying the primary intended users (children or adults) ▪ It does not purport to cover every conceivable hazard of children's jewellery ▪ It does not cover product performance or quality, except as related to safety ▪ This specification has no requirements for those aspects of children's jewellery that present an inherent and recognized hazard as part of the function of jewellery, such as small parts ▪ This specification establishes requirements recognizing that not all jewellery is appropriate for all age groups

6.2. Table 2.a: (Domestic Standards for Raw Materials)

Stage	Code	Scope
Raw Material	**IS 2279: 2018 (Reaffirmed Year: 2019)** - Fine silver bar, sheet, wire, granules and token	▪ This standard covers the requirements of fine silver in the form of bar, sheet, wire, granules and token(does not cover wires for electrical contacts)
	IS 2932: Part 1: 2013 (Reaffirmed Year: 2018) - Enamel, Synthetic, Exterior: (a) Undercoating (b) Finishing Specification : Part 1 for Domestic and Decorative Applications	▪ This standard (Part 1) prescribes requirements, methods of sampling and test for the material commercially known as enamel, synthetic, exterior: (a) undercoating (b) finishing ▪ Colour as required used in painting system for protection and household/decorative purposes

6.3. Table 2.b: (International Standards for Raw Materials)

Stage	Code	Scope
Raw Material	**ISO 9227: 2017** - Corrosion tests in artificial atmospheres — Salt spray tests	▪ Apparatus and Testing methodology ▪ Normative References

Stage	Code	Scope
	GSO ISO 10308: 2013 - Metallic coating	▪ This International Standard reviews published methods for revealing pores (see ISO 2080) and discontinuities in coatings of aluminium, anodized aluminium, brass, cadmium, chromium, cobalt, copper, gold, indium, lead, nickel, nickel-boron, nickel-cobalt, nickel-iron, nickel-phosphorus, palladium, platinum, vitreous or porcelain enamel, rhodium, silver, tin, tin-lead, tin-nickel, tin-zinc, zinc and chromate or phosphate conversion coatings (including associated organic films) on aluminium, beryllium-copper, brass, copper, iron, Nifco alloys, magnesium, nickel, nickel-boron, nickel-phosphorus, phosphor-bronze, silver, steel, tin-nickel and zinc alloy basis metal ▪ The tests summarized in this International Standard are designed to react with the substrate when exposed, by a discontinuity, in such a way as to form an observable reaction product ▪ Pores are usually perpendicular to the coating surface but may be inclined to the coating surface. They are frequently cylindrical in shape but may also assume a twisted shape (see Annex C)

6.4. Table 3.a: (Domestic Standards for Processing)

Stage	Code	Scope
Processing	**IS 964: 1956 (Reaffirmed Year: 2019)** - Methods for chemical analysis of silver solder	▪ This standard covers the methods of chemical analysis of various grades of silver solder. The methods described deal with the determination of silver, copper, cadmium, and zinc
	IS 2113: 2014 (Reaffirmed Year: 2019) - Assaying silver in silver and silver alloys methods	▪ This standard prescribes the methods for determination of silver in silver alloys as below: a) Gravimetric method b) Volumetric (potentiometric) method The methods are suitable for determination of silver in the ranges as specified in IS 2112: 2014 (Reaffirmed Year: 2019) except for fineness 999.9 for which the test method may be agreed to between the parties concerned since their associated impurities have to be determined. The silver alloys may contain copper (Cu), zinc (Zn), cadmium (Cd) and palladium (Pd). These elements except palladium, which must be precipitated before titration, do not interfere with this method of determination of silver ▪ The volumetric method can be used for silver of fineness 99.5 percent and below. These alloys may contain copper, zinc, cadmium and palladium. These elements, except palladium, which must be precipitated before titration, do not interfere with this method of determination of silver. In case of dispute, Gravimetric method may be used as the reference method

Stage	Code	Scope
	IS 1067: 1981 (Reaffirmed Year: 2021) - Electroplated coatings of silver for decorative and protective purposes	▪ This standard specifies requirements of six grades for electroplated coatings of silver for decorative and/or protective applications. It does not apply to coatings on screw threads, to coatings applied on sheet or strip in the fabricated form

6.5. Table 3.b: (International Standards for Processing)

Stage	Code	Scope
Processing	**BS ISO 15096: 2020** - Jewellery and precious Metals-Determination of high purity silver. Difference method using ICP-OES	▪ This document specifies the analytical procedure for the determination of silver with a nominal content of and above 999 ‰ (parts per thousand) ▪ This document specifies a method intended to be used as the recommended method for the determination of silver of fineness of and above 999 ‰. For the determination of fineness of and above 999,9 ‰, modifications described in Annex B apply
	BS EN 1904: 2000 - Precious metals. The fineness's of solders used with precious metal jewellery alloys	▪ Precious metals. The fineness's of solders used with precious metal jewellery alloys
	BS EN ISO 11427: 2016 - Jewellery. Determination of silver in silver jewellery alloys-Volumetric (potentiometric) method using potassium bromide	▪ This International Standard method describes a volumetric method for the determination of silver in jewellery alloys, preferably within the range of fineness stated in ISO 9202 ▪ These alloys may contain copper, zinc, cadmium, and palladium. Apart from palladium, which must be precipitated before commencing titration, these elements do not interfere with this method of determination ▪ This method is intended to be used as the referee method for the determination of fineness in alloys covered by ISO 9202
	AS 1856: 2004 - Electroplated coatings - Silver	▪ This Standard specifies requirements for the electrodeposited coatings of silver used for engineering purposes that may be matt, bright, or semi-bright and are not less than98 percent silver purity ▪ The requirements also specify thickness, quality and testing of electroplated coatings of silver on metallic and non-metallic materials for decorative, protective, electrical contact characteristics, high electrical and thermal conductivity, thermos compression bonding, wear resistance of load-bearing surfaces, spectral reflectivity, solder able surfaces and other general engineering applications
	AS 3895.1-1991 - Methods for the analysis of copper, lead, zinc, gold and silver ores, Part 1: Determination of gold (Fire assay - Flame AAS method)	▪ This Standard sets out a fire assay collection and flame atomic absorption spectrometric finish method for the determination of gold content in copper, lead, zinc, gold and silver ores. The method is applicable to the determination of gold contents from 0.1 g/t to 25 g/t in a range of gold bearing ores

Stage	Code	Scope
	ISO 13756: 2015 - Jewellery - Determination of silver in silver jewellery alloys - Volumetric (potentiometric) method using sodium chloride or potassium chloride	▪ This International Standard specifies a volumetric method for the determination of silver in silver jewellery alloys, preferably within the range of fineness stated in ISO 9202 ▪ These alloys may contain copper, zinc, cadmium, and palladium. Apart from palladium, which must be precipitated before commencing titration, these elements do not interfere with this method of determination

6.6. Table 4.a: (Domestic Standards for Finishing)

Stage	Code	Scope
Finishing	**IS 1417: 2016** - Gold and gold alloys, jewellery/artefacts - Fineness and marking	▪ This standard specifies based on gold content**:** **a.** Grades of fine gold and standard gold used for manufacture of bullion and coins **b.** Grades of gold alloy used in manufacture of jewellery/artefacts ▪ This standard also specifies the guidelines for marking of purity and other details on tested bullion, coins and jewellery/artefacts
	IS 101: Part 4: Section 2: 2021 - Methods of sampling and test for paints, varnishes and related products	▪ Testing of [paint) by visual and spectrometry method ▪ Thickness of coating of paint
	IS 101: Part 4: Section 4: 2020 - Methods of sampling and test for paints, varnishes and related products	▪ Determination of gloss value ▪ Part 4 Optical Test Section 4 Gloss ‘Determination of gloss value at 20Â°, 60Â° and 85Â°’ (Fourth Revision)
	IS 101: Part 5: Section 1: 1988 (Reaffirmed Year: 2019) - Methods of sampling and test for paints, varnishes and related products Part 5 Mechanical test on paint films Sec 1 Hardness tests	▪ Hardness tests for films formed by paints and allied products
	IS 101: Part 8: Section 2: 1990 (Reaffirmed Year: 2017) - Methods of sampling and test for paints, varnishes and related products Part 8 :Tests for pigments and other solids, Sec 2 Pigments and non-volatile matter	▪ Sampling and test for determination of pigment and non-volatile matter content in paints, varnishes and other related products ▪ Adjunct standards and referenced documents
	IS 2112: 2014 (Reaffirmed Year: 2019): Silver and silver alloys, jewellery/artefacts - Fineness and marking - Specification (Third revision)	▪ This standard specifies**:** a. Three grades of fine silver b. Six grades of silver alloys used in the manufacture of jewellery/artefacts of silver based on their silver content ▪ This standard also specifies the guidelines for marking of purity and other details on tested silver jewellery/artefacts

6.7. Table 4.b: (International Standards for Finishing)

Stage	Code	Scope
Finishing	**BS EN ISO 9202: 2019 -** Jewellery and precious metals. Fineness of precious metal alloys	▪ This document specifies a range of fineness of precious metal alloys (excluding solders) recommended for use in the field of jewellery
	ASTM D7786-13 - Standard test method for determining enamel holdout	▪ This standard provides a method for determining the holdout characteristics of a primer and topcoat coating application. A standard topcoat is used to determine the absorption characteristics of a primer. Enamel holdout can be measured as a difference in observed gloss of the topcoat over a primer, relative to the gloss of the same topcoat over a non-porous, smooth surface ▪ The standard is written in the context that the user will be evaluating the enamel holdout characteristics of a primer. Alternatively, the standard may be used as a method to evaluate the enamel holdout characteristics of primer/topcoat system where the primer is constant and different topcoats are used as test paints ▪ This standard may also be used for evaluation of paints other than primers as the first coat. In this alteration the user can test the enamel holdout characteristics of a self-primed topcoat, or use of any other type of paint as the primer followed by the use of a standard topcoat
	ASTM 3359 - Standard test methods for rating adhesion by tape test	▪ Method to evaluate adhesion of a coating to different substrates or surface treatments, or of different coatings to the same substrate ▪ Used to evaluate whether the adhesion of a coating to a substrate is adequate for the user's application ▪ Exclusions
	ASTM D 6659 - Standard practice for xenon-arc exposures of paint and related coatings	▪ Selection of test conditions for accelerated exposure testing of coatings and related products in xenon arc devices conducted according to Practices G151 and G155

Note: The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

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4. Product Information: https://dsource.in/resource/meenakari-art-varanasi/introduction
5. International Legislation/Regulation:
   a. EU: https://echa.europa.eu/regulations/reach/legislation
   b. USA : https://www.compliancegate.com/heavy-metal-regulations-united-states/ https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfcfr/CFRSearch.cfm?CFRPart=111&showFR=1&subpartNode=21:2.0.1.1.11.4
   c. AUS: https://www.compliancegate.com/product-labeling-requirements-australia/
   d. GULF (GCC) : https://www.export.gov/apex/article2?id=Saudi-Arabia-Labeling-Marking-Requirements, https://sabersaudi.co/services/saso-product-coc/saso-technical-regulation /
6. NABL accredited laboratories: https://nabl-india.org/, https://parakh.ncog.gov.in
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8. ISO Standards: https://www.iso.org/obp/ui#search
9. International Standards:
   a. EU : https://www.en-standard.eu/, https://www.bsbedge.com
   b. USA: https://www.en-standard.eu/astm-standards, https://www.astm.org/
   c. AUS : https://www.standards.org.au/
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ITRA (SCENTS), KANNAUJ

1. INTRODUCTION

Kannauj is popularly known as the 'Perfume City' or 'The perfume capital of India'. The district is best known for its essence, flavour and fragrance in the international market.

Attar/Itra, is a traditional Indian perfume manufactured in Kannauj. It has been protected under the Geographical indication (GI) of the Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS) agreement. It is listed at item 157 as "Kannauj Perfume" of the GI Act 1999 of the Government of India with registration confirmed by the Controller General of Patents Designs and Trademarks.

Made from flowers and natural resources. Musk, Camphor, Saffron and other aromatic substances are used for production. Various flowers or varieties may be used depending on the seasons as well, for example:

- **Summer Varieties -** Flower like White Jasmine and plant like vetiver
- **Monsoon Varieties -** Soil known as Mitti Attar that gives the scent of wet earth.
- **Winter Varieties -** Heena attar and musk attar

Itra or Attars are usually obtained through the process of Steam Distillation.

The natural perfume is free of alcohol and chemical except for some productions. Fragrant attar from 'Rose' has more smell while attar that made from 'Sandalwood oil' has lasting smell.

2. RAW MATERIALS

The basic raw materials used are:

2.1. Base Material

Sandal wood oil, Di-octyl Phthalate (DOP) & Liquid paraffin

2.2. Floral Material

Flowers of Gulab, Kewra, Bela, Mehndi, Kadam, Chameli, Marigold, Saffron & Maulshri

2.3. Herb & Spices

A number of herbs and spices are used in this industry which includes Oakmoss, Sugandh mantra, Laurel berry, Juniper berry, Cypriol, Indian valerian, Jatamansi, Hedychium spicatum, Daru Haldi, Sugandha Bala, Sugandha Kokila, Kulanjan, Javitri/Jaiphail, Cardamom, Cloves, Saffron, Ambergris & Musk.

2.4. Sources

Sandal wood oil – South India, Rose, Aligarh (U.P.), Palampur (H.P.), Kewra – Ganjam (Orissa)

3. MANUFACTURING PROCESS

The basic stages involved in manufacturing of 'attar' are:

3.1. Traditional Deg or Still

Degs are made of copper metal and have opening for connections to one or two receivers. Flowers/plant materials are immersed in water within the Deg. The Deg is then sealed using clay.

3.2. Bhapka or Receiver

The Bhapka, also called "receiver" is built of copper and consists of a round shape with long neck. The bhapka is filled with base oils or carrier oils and sealed with strips of cotton cloth and clay. The bhapka and the Deg are connected to each other using a chonga.

Figure 1: Attar Manufacturing

3.3. Traditional Bhatti or Furnace

Traditional Bhatti that are used by attar craftsmen is built up of bricks and clay. Typically, wood or coal is used for heating during the process. The Degs are then heated, which cause the fragrances mixed with steam to move from the Deg to the bhapka.

3.4. Gachchi or cooling water tank

A cooling water tank is the place where Bhapka, or receiver is kept and its purpose is to cooling the distillate from Deg. The steam condenses and the fragrances mix with the base oils and are collected.

3.5. Kuppi or leather bottle

Kuppi are bottles made from leather of animals. These bottles are used to store the attars.

4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- The obtained oils should be clear, devoid of all particulate matter.
- There should be no extraneous matter

5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations (domestic) and international (some key regions/countries) are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

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- Directorate General of Foreign Trade (https://www.dgft.gov.in/CP/)
- Export Promotion Councils (https://commerce.gov.in/useful-links/export-promotion-councils/)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (https://commerce.gov.in/international-trade/trade-agreements/).

5.1. Domestic

S. No.	Country	Regulation	Regulatory Body	Remarks
1	India	Drugs and Cosmetics Act, 1940	Department of Health	Governs the manufacture, sale, import of drugs and cosmetics

5.2. International

S. No.	Region/ Country	Regulation	Regulatory Body	Remarks
1	European Union	a. European Federation of Essential Oils (EFEO) b. International Fragrance Association (IFRA) c. Cosmetic Regulation (EC 1223/2009), EU Commission Regulation (EU) No. 655/2013	a. European Commission b. EFEO, IFRA	a. Published guidance for characterising essential oils b. An essential oil is defined as a volatile part of a natural product, which can be obtained by distillation, steam distillation or expression in the case of citrus fruits
2	United States Of America	a. Food and Drug Administration (FDA)	a. Department of Health and Human Services	a. Governs product requirements for essential oils
3	Australia	a. Australian Register of Therapeutic Goods (ARTG)	a. Department of Health b. Therapeutic Goods Administration (TGA)	a. Essential oil products making therapeutic claims b. Essential oils for use as starting materials
4	Middle East	a. Saudi Standards, Metrology and Quality Organization (SASO) regulations	a. Saudi Standards, Metrology and Quality Organization (SASO)	a. General product safety regulation b. Labelling requirement

6. VOLUNTARY STANDARDS

Domestic Standards			International Standards		
Stage	**Code**	**Stage**	**Region/ Country**	**Standard**	**Aspects Covered**
General (Refer Table 1)	**IS 6597: 2001 (Reaffirmed Year: 2017)**	Glossary of Terms Relating to Fragrance and Flavour Industry	-	-	-
Raw Materials (Refer Table 2.a & 2.b)	**IS 2284: 1988 (Reaffirmed Year: 2019)**	Method for olfactory assessment of natural and synthetic perfumery materials	Australia	**AS 4550**	Essential oils - Sampling
	-	-	Middle East	**GSO/ISO 212: 2008**	Essential oils - Sampling
Manufacturing/ Processing (Refer Table 3.a & 3.b)	**IS 326: Part 1: 2018**	Methods of sampling and test for natural and synthetic perfumery materials	USA	**ISO/TC 54**	Standardization of methods of analysis and specifications for essential oils.
	IS 15740: 2007 (Reaffirmed Year: 2017)	Oil of Rose - Specification	USA	**ICS 71.100.60**	Complete list of Standard of essential oils
	IS 327: 1991(Reaffirmed Year: 2019)	Oil of Lemongrass - Specification	Australia	**CH-021**	Complete list of Standard of essential oils
	IS 328: 2016 (Reaffirmed Year: 2021)	Oil of Eucalyptus Globulus - Specification	Australia	**AS 5025**	Essential oils — General guidance on chromatographic profiles
	IS 329: 2004 (Reaffirmed Year: 2020)	Oil of Sandalwood – Specification	Middle East	**GSO ISO 11024-1: 2010**	Essential oils — General guidance on chromatographic profiles
	IS 512: 1988 (Reaffirmed Year: 2019)	Oil of Citronella (Java) – Specification	-	-	-
Finishing (Refer Table 4)	-	-	Middle East	**GSO ISO/TS 210: 2021**	Essential oils — General rules for packaging, conditioning and storage

6.1. Table 1: (Domestic & International General Standards)

Stage	Code	Scope
General	**IS 6597: 2001 (Reaffirmed Year: 2017)** - Glossary of Terms Relating to Fragrance and Flavour Industry	▪ Defines the terms relating to fragrance and flavour industries comprising of natural ingredients (essential oils, oleoresin, balsams, resinoids, absolutes) and synthetic aroma chemicals for producing finished fragrance and flavours composition

6.2. Table 2.a: (Domestic Standards for Raw Materials)

Stage	Code	Scope
Raw Material	**IS 2284: 1988 (Reaffirmed Year: 2019)** - Method for olfactory assessment of natural and synthetic perfumery materials	▪ prescribes method for olfactory assessment of natural and synthetic perfumery materials based on comparison of a given material with its corresponding standard sample

6.3. Table 2.b: (International Standards for Raw Materials)

Stage	Code	Scope
Raw Material	**AS 4550** - Essential oils Sampling	▪ Gives the general rules for the sampling of essential oils, in order to provide a laboratory with quantities that are suitable to be handled for expertise purposes
	GSO/ISO 212: 2008 - Essential oils Sampling	▪ Gives the general rules for the sampling of essential oils, in order to provide a laboratory with quantities that are suitable to be handled for expertise purposes

6.4. Table 3.a: (Domestic Standards for Processing)

Stage	Code	Scope
Processing	**IS 326: Part 1: 2018** -Methods of sampling and test for natural and synthetic perfumery materials	▪ Methods of sampling for natural and synthetic perfumery materials, for the purpose of determining their organoleptic, physical and chemical characteristics
	IS 15740: 2007 (Reaffirmed Year: 2017) - Oil of Rose - Specification	▪ Specifies the requirements and the methods of sampling of essential oil of rose (Rosa damascenaMiller)
	IS 327: 1991(Reaffirmed Year: 2019) - Oil of Lemongrass - Specification	▪ Prescribes the requirements and the methods of test for the material commercially known as the oil of lemongrass or the East Indian Oil of lemongrass ▪ The material is largely used in the extraction of citral, the chief constituent of the oil and the starting material for the manufacture of important ion-ones. Itis also employed in the preparation of lemon-like perfumes, germicides, etc.
	IS 328: 2016(Reaffirmed Year: 2021) - Oil of Eucalyptus Globulus - Specification	▪ Prescribes the requirements and the methods of sampling and test for oil of Eucalyptus globulus

Stage	Code	Scope
	IS 329: 2004 (Reaffirmed Year: 2020) - Oil of Sandalwood – Specification	▪ Prescribes the requirements and the methods of sampling and test for oil of sandalwood
	IS 512: 1988 (Reaffirmed Year: 2019) : Oil of Citronella (Java) – Specification	▪ Prescribes the requirements and the method of sampling and test for oil of citronella (Java)

6.5. Table 3.b: (International Standards for Processing)

Stage	Code	Scope
Processing	**ISO/TC 54** - specifications for essential oils	▪ Standardization of methods of analysis and specifications for essential oils
	ICS 71.100.60 - Complete list of Standard of essential oils	▪ Gives a list of the botanical names of plants used for the production of essential oils, together with the common names of the essential oils ▪ An alphabetical index of common names of essential oils is included at the end of this document
	CH-021 - Complete list of Standard of essential oils	▪ Gives a list of the botanical names of plants used for the production of essential oils, together with the common names of the essential oils
	AS 5025 - Essential oils — General guidance on chromatographic profiles	▪ Describes general guidelines on the determination of the chromatographic profile of an essential oil by gas chromatography on a capillary column ▪ The chromatographic profile is one of the specifications which enables assessment of the quality of an essential oil in the same way as the physico-chemical characteristics. It is determined at the time of finalizing the standard on the essential oil ▪ It is not a determination of the true concentration of the components, it is only an evaluation of its relative proportions
	GSO ISO 11024-1: 2010 - Essential oils — General guidance on chromatographic profiles	▪ Describes general guidelines on the determination of the chromatographic profile of an essential oil by gas chromatography on a capillary column ▪ The chromatographic profile is one of the specifications which enables assessment of the quality of an essential oil in the same way as the physico-chemical characteristics. It is determined at the time of finalizing the standard on the essential oil ▪ It is not a determination of the true concentration of the components, it is only an evaluation of its relative proportions

6.6. Table 4: (International Standards for Finishing)

Stage	Code	Scope
Finishing	**GSO ISO/TS 210: 2021** - Essential oils General rules for packaging, conditioning and storage	▪ Describes the specifications to be met by the containers intended for containing essential oils, as well as recommendations relating to their conditioning and storage ▪ Essential oils are used for different purposes, as follows: **a.** ¾ food use; **b.** ¾ pharmaceutical use; **c.** ¾ perfumery and cosmetic use; **d.** ¾ reference samples or test samples;

Stage	Code	Scope
		e. ¾ industrial raw materials. ▪ According to the use of the essential oils, it is necessary to use appropriate containers which also meet the requirements of national, European or international regulations

Note: The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

REFERENCES

1. Product information: http://odopup.in/en/article/Kannauj
2. Product information: https://en.gaonconnection.com/the-history-of-fragrance-how-attar-or-ittar-has-evolved-from-one-era-to-the-other/ https://www.kannaujattar.com/blog/ittar-history-usage-benefits-how-to-apply/ http://ijirssc.in/pdf/1464791349.pdf
3. Product Information: https://kannauj.city/kannauj-attar-and-its-traditional-manufacturing-method/
4. Raw material & manufacturing - https://www.kannaujattar.com/blog/ittar-history-usage-benefits-how-to-apply/ , http://ijirssc.in/pdf/1464791349.pdf
5. International Legislation/Regulation:
   a. Europe - https://echa.europa.eu/support/substance-identification/sector-specific-support-for-substance-identification/essential-oils , https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32013R0655
   b. America - https://www.fda.gov/cosmetics/cosmetic-products/aromatherapy
   c. Australia - https://www.tga.gov.au/sites/default/files/cm-argcm-p4.pdf
6. NABL accredited laboratories: https://nabl-india.org/ , https://parakh.ncog.gov.in
7. Domestic Standards: https://www.services.bis.gov.in:8071/php/BIS/PublishStandards/published/standards?commttid=NTQ%3D
8. International Standards:
   a. USA - https://www.iso.org/standard/66365.html
   b. Australia – https://www.standards.org.au/standards-catalogue/sa-snz/health/ch-021?page=4
   c. Gulf nation - https://members.wto.org/crnattachments/2013/sps/SAU/13_4023_00_e.pdf

APPENDIX I: TESTING AND CERTIFICATION

Uttar Pradesh is home to a large number of indigenous products and manufacturing units associated with these unique products. The crafts and artisans have both withstood the vagaries of time and have flourished under the patronage provided in the form of various schemes to promote these products and bring about innovation in the industry by up skilling the artisans as well as providing them assistance in tapping new markets and becoming competitive in the modern market by utilizing machinery developed locally to augment their efforts.

With the advent of machinery and technology there has been a gradual move towards standardization to improve both the product life cycle as well as the quality of materials used. Two important aspects in standardization of products and processes are:

- Testing (Product/Material testing)
- Certification (Product Certification or Systems Certification)

1. PRODUCT TESTING

What is product testing?

Testing is the determination of one or more of a product's characteristics and is usually performed by a laboratory. In other words, product testing is a process to measure its performance, safety, quality, and compliance with established standards.

Quality testing is involved at each step of the production process and often begins with the testing of raw materials, moving through the testing of in-process products to the finished product.

Testing is an essential part of developing a quality product. Any product that is produced is expected to meet certain requirements and specifications to ensure standardization and consistency.

Why is testing of products required?

Testing is an essential part of developing a quality product. Product testing is required to assure quality, performance and safety of the product. Product testing is also required to comply with the regulatory requirements and standards, wherever prescribed.

Testing at the various stages of production helps to ensure that the products meet the defined specifications and gain acceptance globally.

Why should you get the product testing done from an approved/ accredited laboratory?

It is always recommended to get the tests done at a Laboratory that is approved /accredited by a competent authority. Accreditation indicates a formal recognition by an independent body, generally known as an accreditation body, which a certification body operates according to international standards.

Accreditation is an international network which is managed by the International Accreditation Forum (IAF), in the fields of management systems, products, services, personnel and other similar programmes of conformity assessment, and the International Laboratory Accreditation Cooperation (ILAC), in the field of laboratory and inspection accreditation. ILAC and IAF are international associations of accreditation bodies and they facilitate world trade through mutual recognition and also enhances the acceptance of products and services across national borders. Details on these international bodies and their MRA framework are available on www.iaf.nu and www.ilac.org, respectively.

In India, the National Accreditation Board for Testing and Calibration Laboratories (NABL), a constituent Board of Quality Council of India (QCI) has been established as the National Accreditation Body for accreditation of such laboratory. NABL is Mutual Recognition Arrangements (MRA) signatory to ILAC as well as APAC for the accreditation of Testing and Calibration Laboratories (ISO/IEC 17025), Medical Testing Laboratories (ISO 15189), Proficiency Testing Providers (PTP) (ISO/IEC 17043) and Reference materials producers (RMP).

Such MRA reduces technical barrier to trade and facilitates acceptance of test/ calibration results between countries which MRA partners represent.

Further, for certain products like food, pharmaceuticals etc., there are regulatory requirements (domestic as well as region or country specific (if exporting)) for testing of the raw materials and finished goods which are mandatory to be followed. For such products, various regulations and the standards making bodies like Bureau of Indian Standards (BIS) also notify specific Laboratories, from time to time. For e.g. BIS has recognised several laboratories for the tests prescribed under the Indian Standards formulated by them. Similarly, other regions or countries may have specific regulations that require prescribed tests, for e.g. REACH regulation of the European Union, specifies requirements to for the protection of human health and the environment from the risks that can be posed by chemicals.

Where can you get the list of accredited/approved laboratories?

List of the testing laboratories for the relevant tests may be obtained from the following URLs:

- **National Accreditation Board for Testing and Calibration Laboratories (NABL)**: https://nabl-Indiaorg/
- **Bureau of Indian Standards (BIS)**: https://bis.gov.in/

2. CERTIFICATION

Certification is a provision by an independent body of written assurance (a certificate) that the product, service or system in question meets specific requirements.

Broadly, certifications may be categorised as:

a. Management Systems Certification

These certifications help organizations improve their performance by specifying repeatable steps that organizations consciously implement to achieve their goals and objectives, and to create an organizational culture that reflexively engages in a continuous cycle of self-evaluation, correction and improvement of operations and processes through heightened employee awareness and management leadership and commitment.

The benefits of an effective management system to an organization include:

- More efficient use of resources and improved financial performance
- Improved risk management and protection of people and the environment
- Increased capability to deliver consistent and improved services and products, thereby increasing value to customers and all other stakeholders

Examples of Systems Certification include ISO 9001 QMS, ISO 22000 FSMS, etc.

b. Product Certification

Product certification is the process of certifying that a specific product fulfils requirements prescribed in contracts, regulations or specifications and has passed all necessary quality assurance and performance tests. Product certification is often required in industries and marketplaces where product failure could bring serious negative consequences to the health and safety of people and environment.

The benefits of an effective management system to an organization include:

- Product certification gives product users the confidence they need to purchase the desired products.
- If a product is certified according to applicable national or international laws, customers know that that product will function correctly and won't cause any harm. Examples of product certification include CE marking, ISI mark, Hallmark, etc.

Are all certifications mandatory?

Mostly, the certifications are voluntary in nature. However, for a number of products, compliance to specific Standards may be made compulsory, for instance, compliance to Indian Standards for a number of products, has been made compulsory by the Central Government under various considerations viz. public interest, protection of human, animal or plant health, safety of environment, prevention of unfair trade practices and national security. For such products, the Central Government directs mandatory use of Standard Mark under a Licence or Certificate of Conformity (CoC) from BIS through issuance of Quality Control Orders (QCOs). Similarly, other regions or countries may have specific regulations that require prescribed certifications, for e.g. USFDA prescribes specific requirements for Food, Drugs and Cosmetic products.

Why should you get the Certification from an accredited Certification Body?

It is always recommended to obtain the certification from accredited Certification Bodies or the ones that are prescribed by the relevant regulatory authorities. Accreditation indicates a formal recognition by an independent body, generally known as an accreditation body, which a certification body operates according to international standards.

Accreditation is an international network which is managed by the International Accreditation Forum (IAF), in the fields of management systems, products, services, personnel and other similar programmes of conformity assessment, and the International Laboratory Accreditation Cooperation (ILAC), in the field of laboratory and inspection accreditation. ILAC and IAF are international associations of accreditation bodies and they facilitate world trade through mutual recognition and also enhances the acceptance of products and services across national borders. Details on these international bodies and their MRA framework are available on www.iaf.nu and www.ilac.org, respectively.

In India, the National Accreditation Board for Certification Bodies (NABCB), a constituent board of Quality Council of India (QCI) has been established as the National Accreditation Body for accreditation of such Certification and Inspection Bodies. The National Accreditation Board for Certification Bodies provides accreditation to Certification and Inspection Bodies based on assessment of their competence as per the Board's criteria and in accordance with International Standards and Guidelines.

NABCB is internationally recognized and represents the interests of the Indian industry at international forums through membership and active participation with the objective of becoming a signatory to international Multilateral / Mutual Recognition Arrangements (MLA / MRA).

Where can you get the list of accredited/approved laboratories?

List of the NABCB accredited Certification Bodies (CBs) and Inspection Bodies (IBs) for relevant certifications may be obtained from the following URL:

National Accreditation Board for Certification Bodies (NABCB): http://nabcb.qci.org.in/

APPENDIX II: IMPORTANT INSTITUTIONS

MISCELLANEOUS				
S. No.	**Institute**	**Areas of Expertise**	**Postal Address**	**Contact**
1	Fragrance & Flavour Development Centre (FFDC)	▪ Training & Consultancy ▪ Agro-Tech ▪ Supply of Planting materials, Seeds ▪ Processing on Job work basis ▪ Creation of Fragrances	Fragrance & Flavour Development Centre (FFDC) Ministry of MSME, Govt. of India. T. Road, Makrand Nagar, Kannauj – 209726 Uttar Pradesh	Email: ffdcknj@gmail.com Phone: 9415334050 Website: www.ffdcindia.org
2	CSIR-North East Institute of Science & Technology	▪ Processes and technology development (Essential oils, Incense, Speciality Papers) ▪ Transfer of Technology	CSIR-North East Institute of Science & Technology (Formerly Regional Research Laboratory), Jorhat-785006, Assam	Email: director@neist.res.in Phone: +91 376-2370011 Website: www.neist.res.in

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